

Lecture 12: Empirical Bayes

*Instructor: Jonathan Pipping-Gamón**Authors: JP, RB*

12.1 Which Batting Averages Will Persist?

At the end of the 2020 Covid-shortened MLB season, some players finished with unusually high or low batting averages in relatively few at-bats. Suppose our job is to use those 2020 results to predict 2021.

For player i , let

$$X_i = \frac{H_i}{N_i}$$

be the observed batting average from H_i hits in N_i at-bats, and let μ_i be the player's underlying batting ability.

The raw estimate is

$$\hat{\mu}_i^{\text{MLE}} = X_i.$$

But the leaderboard mixes real differences in player quality with sampling noise. A .390 average in 40 at-bats should move more than a .310 average in 300 at-bats.

When no external prior is available, the observed group of related players can supply information about a reasonable league distribution. Empirical Bayes uses that distribution as the shrinkage target, so the dataset itself helps decide both where noisy estimates are pulled and how strongly they move.

12.2 Pooling Strategies

There are three broad strategies for estimating many related quantities:

- **No pooling:** estimate every player separately using X_i .
- **Complete pooling:** predict every player with one league mean.
- **Partial pooling:** move every player toward the league mean, but not all the way.

Partial pooling recognizes that players differ, while also recognizing that every player in this dataset belongs to a related league population.

12.3 Beta–Binomial Partial Pooling

A convenient partial-pooling model for hits is

$$\begin{aligned} H_i \mid p_i &\sim \text{Binomial}(N_i, p_i), \\ p_i &\sim \text{Beta}(\alpha, \beta). \end{aligned}$$

Then

$$\mathbb{E}[p_i \mid H_i, N_i] = \frac{H_i + \alpha}{N_i + \alpha + \beta}.$$

Empirical Bayes estimates (α, β) from the league rather than choosing them externally. The Normal approximation used next makes the amount of shrinkage and the sources of variation especially transparent.

Example 12.1 (A Hot Start in 20 At-Bats). Suppose a player has $H_i = 9$ hits in $N_i = 20$ at-bats, and the estimated league distribution is $\text{Beta}(60, 180)$. The raw batting average is $9/20 = 0.450$, while

$$\hat{p}_i^{\text{EB}} = \frac{9 + 60}{20 + 60 + 180} = 0.265.$$

The short hot streak moves the estimate above the prior center of 0.250, but 20 at-bats provide little evidence for a true .450 hitter.

12.4 Normal–Normal Partial Pooling

A convenient approximation is

$$\begin{aligned} X_i \mid \mu_i &\sim \mathcal{N}(\mu_i, \sigma_i^2), \\ \mu_i &\sim \mathcal{N}(\mu, \tau^2). \end{aligned}$$

The first line describes sampling noise for player i . For a batting average,

$$\sigma_i^2 \approx \frac{C}{N_i},$$

so players with fewer at-bats have noisier observed rates. The second line describes genuine differences across players:

- μ is the population center.
- τ^2 is the between-player variance in underlying ability.

The posterior mean is

$$\begin{aligned} \mathbb{E}[\mu_i \mid X_i] &= \frac{\tau^2}{\tau^2 + \sigma_i^2} X_i + \frac{\sigma_i^2}{\tau^2 + \sigma_i^2} \mu \\ &= \mu + \lambda_i (X_i - \mu), \end{aligned}$$

where

$$\lambda_i = \frac{\tau^2}{\tau^2 + \sigma_i^2}.$$

12.5 What Determines Shrinkage?

The posterior mean balances two sources of information: the player's observed batting average and the population mean. The weight λ_i determines how much of player i 's observed difference from the league mean survives:

- Large N_i means small σ_i^2 , so the player's own average is more reliable and $\lambda_i \approx 1$.
- Small N_i means large σ_i^2 , so the player's own average is noisy and the estimate pools more strongly.
- Large τ^2 means players genuinely differ, so population-level information is less decisive and shrinkage is weaker.
- Small τ^2 means players are more similar, so the population mean is more informative and shrinkage is stronger.

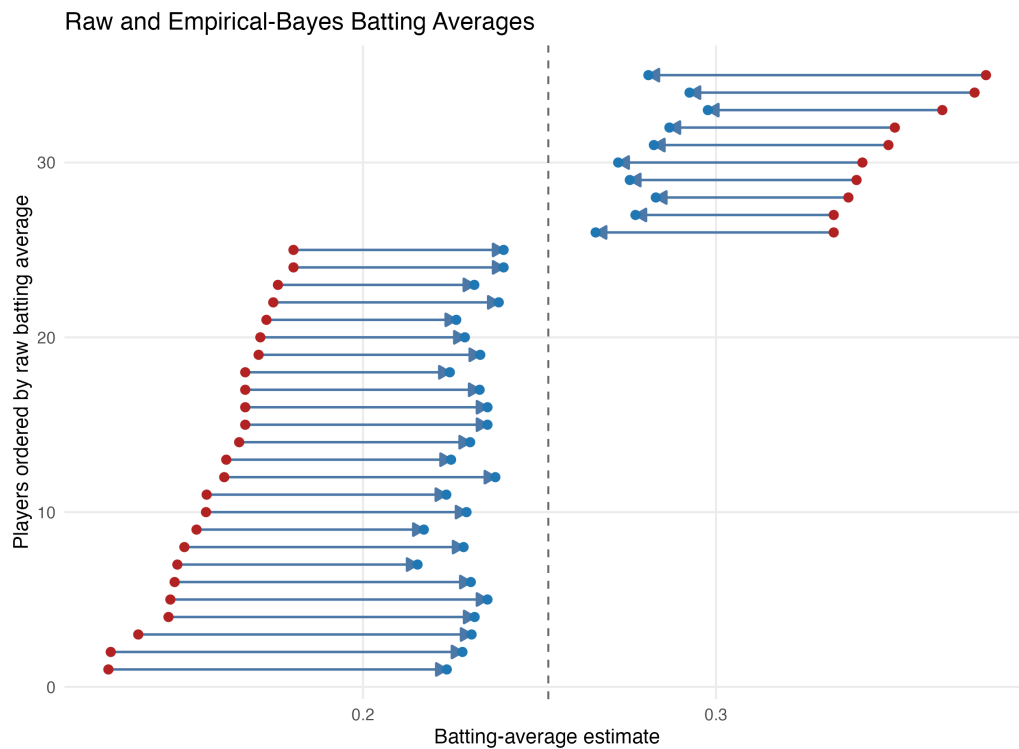


Figure 12.1: Raw and empirical-Bayes estimates for the 35 largest shrinkage adjustments.

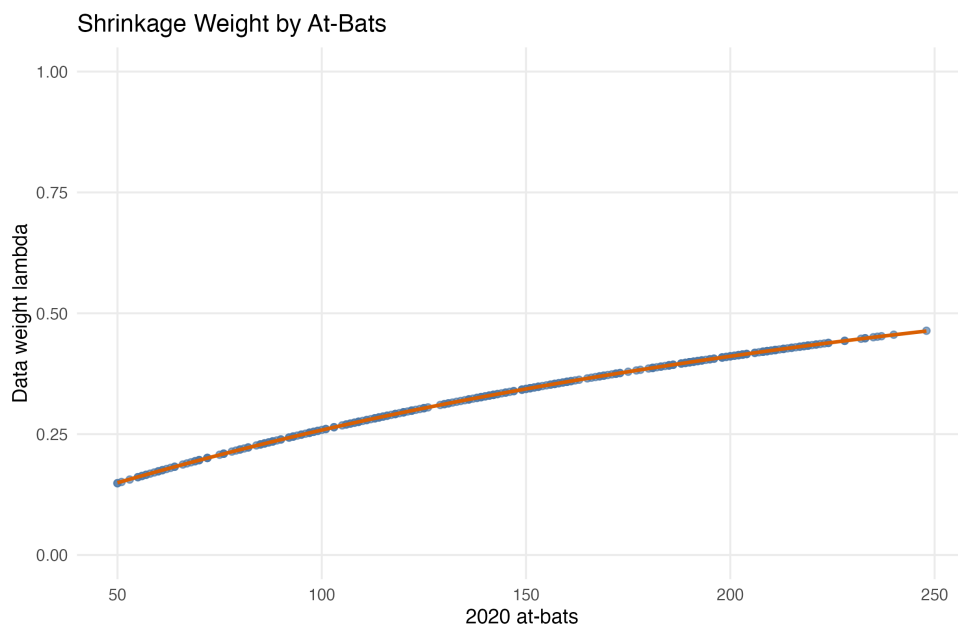


Figure 12.2: Empirical-Bayes data weight by 2020 at-bats.

12.6 Estimating the Prior From the League

The posterior mean assumes that μ and τ^2 are known. In practice, they are not. This is where Empirical Bayes comes in.

Definition 12.2 (Empirical Bayes). *Empirical Bayes estimates the hyperparameters of a prior distribution from the observed population, then plugs those estimates into a Bayesian posterior calculation.*

After estimating the hyperparameters, ordinary empirical Bayes treats them as *fixed*. Player-level uncertainty is then conditional on one fitted league distribution, $(\hat{\mu}, \hat{\tau}^2)$. A fully Bayesian model would also propagate uncertainty about (μ, τ^2) , so empirical-Bayes intervals can be too narrow, especially with a small or noisy population.

To estimate the league distribution, empirical Bayes uses the marginal distribution of the observed batting average. The unobserved ability μ_i is integrated out:

$$\begin{aligned} p(X_i | \mu, \tau^2) &= \int p(X_i | \mu_i) p(\mu_i | \mu, \tau^2) d\mu_i \\ &= \mathcal{N}(X_i; \mu, \tau^2 + \sigma_i^2). \end{aligned}$$

The variance separates two sources of leaderboard spread:

$$\text{observed variation} = \text{real between-player variation} + \text{sampling noise}.$$

Assuming players are conditionally independent, the marginal likelihood multiplies their marginal densities. Taking logs turns that product into a sum, so the empirical-Bayes hyperparameter estimates solve

$$(\hat{\mu}, \hat{\tau}^2) = \arg \max_{\mu, \tau^2 \geq 0} \sum_i \log \mathcal{N}(X_i; \mu, \tau^2 + \sigma_i^2).$$

This chooses the league mean and between-player variance that make the observed batting averages most plausible, after accounting for the fact that some players had noisier 2020 averages than others.

A simpler method-of-moments estimate uses

$$\begin{aligned}\hat{\mu} &\approx \text{league batting average,} \\ \hat{\tau}^2 &\approx \max \{0, \text{Var}(X_i) - \text{Mean}(\sigma_i^2)\}.\end{aligned}$$

The empirical-Bayes estimate is then

$$\hat{\mu}_i^{\text{EB}} = \hat{\mu} + \frac{\hat{\tau}^2}{\hat{\tau}^2 + \sigma_i^2}(X_i - \hat{\mu}).$$

The prior is not an external scouting report here. The prior is the estimated league distribution.

12.7 Out-of-Sample Evaluation

A shrunk leaderboard may look more plausible, but future prediction tests whether the estimator improves the intended task.

Here, batting averages from the shortened 2020 season are used to predict 2021 batting averages. The 2021 data are not used to fit the empirical-Bayes model.

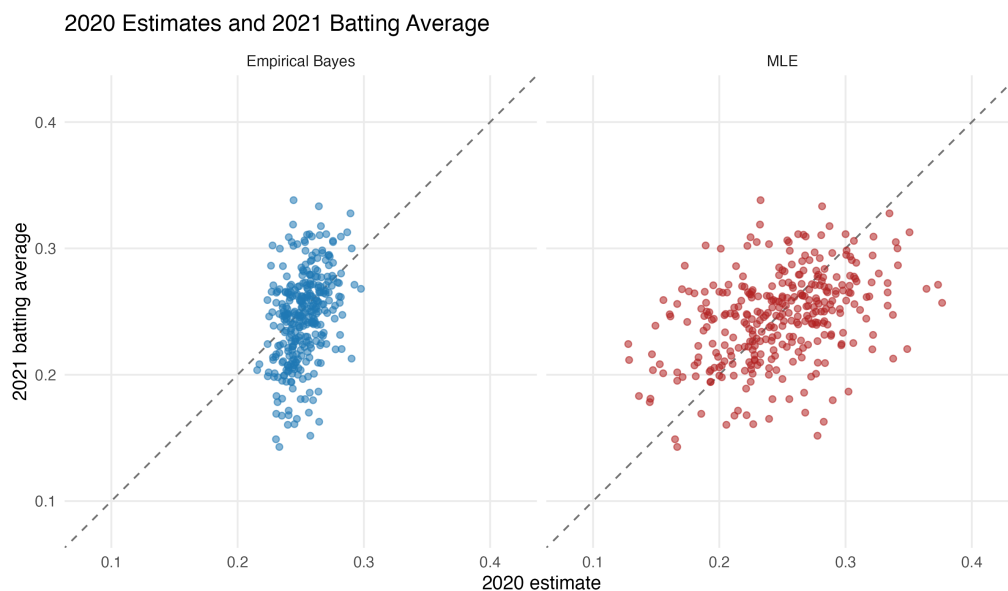


Figure 12.3: Raw and empirical-Bayes 2020 estimates against 2021 batting average.

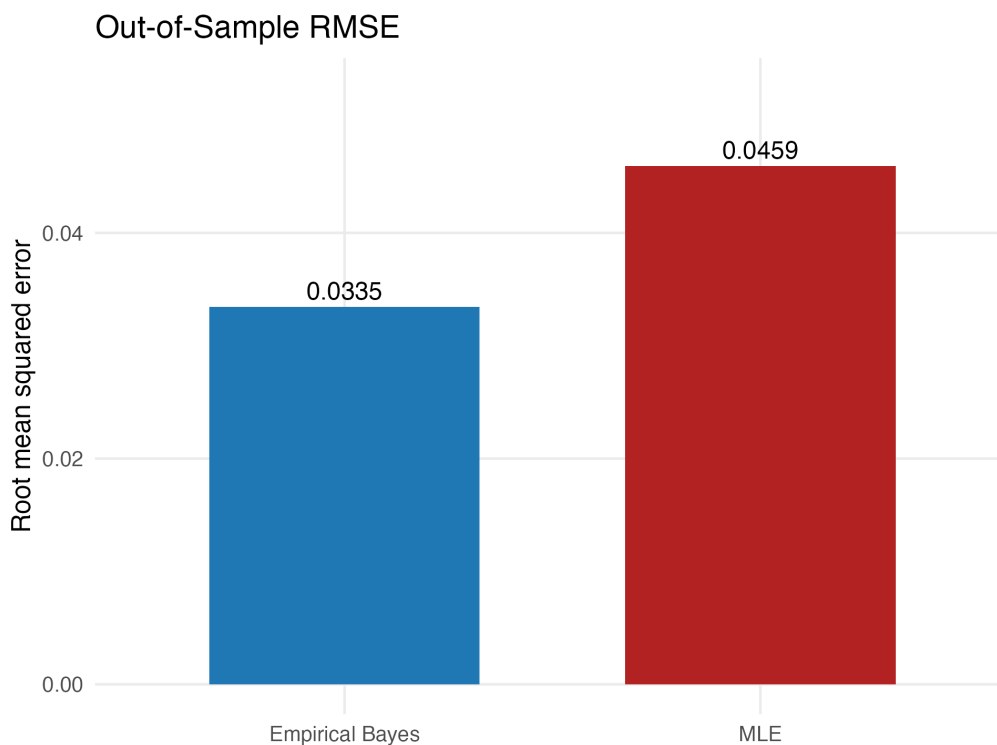


Figure 12.4: Out-of-sample RMSE against 2021 batting average.

12.8 When Should Players Share Information?

Empirical Bayes borrows strength by pooling information across related players. The full group estimates the prior center $\hat{\mu}$ and spread $\hat{\tau}^2$; those estimates determine where each noisy batting average is pulled and how strongly it moves. This only makes sense when players are comparable enough to share a population model. However, this exchangeability assumption can fail:

- pitchers and position players may belong to different populations;
- rookies and veterans may need different prior centers;
- park, injury, role, and aging information may explain real heterogeneity.

The model is where we decide which players should share information. For example, instead of shrinking every player toward one common league mean, we could set up

$$\mu_i \sim \mathcal{N}(z_i^T \gamma, \tau^2),$$

where z_i contains characteristics such as position, experience, or park context. Players with similar z_i shrink toward similar targets, while τ^2 captures remaining differences not explained by those characteristics.

The same structure applies to free-throw rates, three-point shooting, quarterback efficiency, team scoring rates, and prospect projections. Partial pooling is extremely useful for both prediction and inference when related quantities are measured with substantial noise.

12.9 Takeaways

- A leaderboard contains both signal and sampling noise.
- Partial pooling balances each unit's data against information from the population.
- Empirical Bayes estimates the population distribution from the observed groups.
- Shrinkage is stronger for noisier observations and weaker when groups genuinely differ.
- Out-of-sample prediction and exchangeability checks determine whether the pooling strategy is useful.

Appendix

A Normal–Normal Posterior Derivation

Combining the Normal likelihood and Normal prior gives

$$p(\mu_i \mid X_i) \propto \exp \left\{ -\frac{(X_i - \mu_i)^2}{2\sigma_i^2} - \frac{(\mu_i - \mu)^2}{2\tau^2} \right\}.$$

Collecting the terms involving μ_i produces a Normal posterior with precision $1/\sigma_i^2 + 1/\tau^2$ and mean

$$\frac{X_i/\sigma_i^2 + \mu/\tau^2}{1/\sigma_i^2 + 1/\tau^2} = \mu + \frac{\tau^2}{\tau^2 + \sigma_i^2}(X_i - \mu).$$

B Suggested Reading

- Bradley Efron and Carl Morris, “Stein’s Paradox in Statistics,” for the relationship between empirical Bayes and shrinkage.
- Andrew Gelman and Jennifer Hill, *Data Analysis Using Regression and Multilevel/Hierarchical Models*, on partial pooling.